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APPLICATION OF FLEXIBLE STAYBOLTS AND EXPANSION RADIAL STAYS (Cont'd).

The sketches shown on page 290 illustrate application of expansion radial stays and flexible staybolts in what is generally considered as the breaking zones of locomotive fireboxes. Fig. 1 shows fireboxes without combustion chambers and Fig. #2 shows fireboxes with combustion chambers.

Expansion radial stays are to be applied only in first four rows back of rear flue sheet except on classes having combustion chamber which have complete installation of expansion stays over combustion chamber and crown sheet as shown in Fig. #2.

The sketches shown on page 290 are not exactly correct for all classes of locomotives. The drawings listed in table must be followed when applying expansion radial stays and flexible staybolts.

When renewal of crown sheet is necessary the four rows of expansion radial stays are to be introduced on classes listed below having original crown bar arrangement.

Flexible staybolts are to be applied when new firebox or new crown sheet is applied or when the locomotive is shopped for general or heavy repairs provided the box is good for at least four years additional service. This refers to engines not previously equipped with flexible staybolts or stays.

The following table shows classes authorized to receive expansion radial stays and flexible staybolts.

Class	Drwg. No.	Remarks	Class	Drwg. No.	Remarks	Class	Drwg. No.	Remarks	Class	Drwg. No.	Remarks	Class	Drwg. No.	Remarks	Class	Drwg. No.	Remarks
542	140/140		885	140/217		1270	140/135		1452	140/140		1950	140/209		3450		
564	37/212		900	140/201		1290	140/135		1480	37/203		3010	37/215		3500	140/206	
566	140/260		990	140/139		1297	37/200		1550	140/140			140/20_		#3700	37/225	See Note
+729	13/179		1000	37/217	140/243		37/203		1600	140/201		3100	140/205		3751		
+759	13/179		1014	37/217	37/217	1309	37/200		1674	140/200		3129	140/220		#3800		See Note
+769	13/179		1050	37/217		1337	140/218		1798	37/202			37/219		4000		
789	140/307		1200	140/216		1398	37/214		1800	140/208		3160			4101		
825	140/307		1226	140/216		1400	140/140		1900	140/207		3400					

\+ Flexible staybolts and flexible radial stays applied when new back end is applied.

\# Some locomotive numbered between Nos. 3700 to 3736 and 3800 to 3849 were originally built with partial installation of flexible stays over the crown sheet. As engines 3800 to 3849 west of Albuquerque and all engines 3700 to 3736 pass through the shops they are to receive complete installation of flexible stays over the entire crown sheet including the combustion chamber. Engines 3800 to 3849 running east of Albuquerque will continue original solid stays until inspection of radial stays justifies the removal due to breakage or at such times these engines are transferred to other territory.

Engines 3737 to 3750 inclusive, and 3850 to 3940 inclusive, were built with complete installation of flexible stays over crown sheet.

The following engines are equipped with flexible staybolts and radial stays having reduced centers as shown on page 16b, 292-A and 292-B.

Class	Engine Nos.	Class	Engine Nos.	Class	Engine Nos.	Class	Engine Nos.
2900		3700	3737-3750	3800	3850-3940	5001	
3400	3430-3449	3751		4000	4015-4100	5011	
3450		3765		4101			
3460		3776		5000			

All other engines equipped with flexible staybolts and radial stays having centers of same diameter.

The expansion and flexible staybolts on following locomotives are hollow and arranged for electric testing. Classes 2900, 3450, 3460, 3751, 3765, 3776, engines 3891-3940, engines 4086-4100 and classes 4101, 5000, 5001 and 5011.

Hollow flexible stays are to be cleaned and tested at each hydrostatic test of the boiler, as specified in I.C.C. Rule 23-b. Caps are not to be removed for two year inspection from outer ends of hollow flexible bolts if same have been regularly and properly cleaned and tested in accordance with I.C.C. Rule 23-b.

When a locomotive having radial stayed firebox with front of crown sheet supported by Tee bars and sling stays is in shop for general repairs and the Tee bars have to be removed for any reason such as defective stays, patching crown sheet, etc., the sling stay arrangement is to be discarded and replaced by expansion stays.